TABASUM RAHNUMA

POSTDOCTORAL RESEARCHER · THEORETICAL PHYSICS

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• tabasum-ph.github.io

My research lies at the intersection of quantum gravity and scattering amplitudes, focusing on the asymptotic structure of spacetime and its connections to gravitational wave physics. I study how symmetries at the boundaries of spacetime — BMS and its extensions — encode infrared physics through soft theorems and memory effects. Recently I have developed recursive post-Minkowskian methods to construct gravitational metrics perturbatively, with applications to gravitational wave observables.

RESEARCH INTERESTS

Quantum Gravity • Asymptotic Symmetries • General Relativity • Scattering Amplitudes • Celestial Holography • Gravitational Wave Physics • Post-Minkowskian Formalisms • Flat Space Holography

COMPUTATIONAL SKILLS

Mathematica (xAct, xTensor tensor algebra packages), ROOT/RooFit (data analysis), Fortran 77, C, C++.

POSITIONS & EDUCATION

Nov 2025 – present
QUC Research Fellow
Korea Institute for Advanced Studies (KIAS), Seoul, South Korea
Research: Quantum Gravity, Asymptotic Symmetries, Scattering Amplitudes, Gravitational Wave Physics.

Oct 2024 – Oct 2025
Postdoctoral Researcher
Asia Pacific Center for Theoretical Physics (APCTP), Pohang, South Korea
Junior Research Group: Dualities in String/M-theory and Quantum Gravity.

Jul 2019 – Jul 2024
Doctor of Philosophy in Physics
Indian Institute of Science Education and Research Bhopal (IISERB), India
Thesis: "Scattering in Asymptotically Flat Spacetimes and Symmetries from Holography"
Supervisor: Dr. Nabamita Banerjee

Jul 2016 – Apr 2018
Master of Science in Physics
Utkal University, Bhubaneswar, Odisha, India
Specialisation: Advanced Particle Physics and Field Theory. Programming: Fortran 77, C, C++.

Jul 2012 – Apr 2015
Bachelor of Science in Physics (Hons.)
Gopabandhu Science College, Cuttack, Odisha, India

PUBLICATIONS

6 · 2026
Iterative Solution of the Kerr Black Hole Metric
P. H. Damgaard, H. Lee, K. Lee, T. Rahnuma
arXiv: 2605.19948 [hep-th, gr-qc]

5 · 2026
Gravitational Metric of a Star
P. H. Damgaard, H. Lee, K. Lee, T. Rahnuma
arXiv: 2603.16493 [hep-th, gr-qc]

4 · 2025
AdS S-Matrix for Massive Vector Fields
N. Banerjee, A. N. Desai, K. Fernandes, A. Mitra, T. Rahnuma
JHEP 10.1007/JHEP05(2025)094

3 · 2024
Asymptotic Symmetry Algebra of $\mathcal{N}=8$ Supergravity
N. Banerjee, T. Rahnuma, R. K. Singh
Physical Review D 10.1103/PhysRevD.109.046010

2 · 2023
Soft and Collinear Limits in $\mathcal{N}=8$ Supergravity using Double Copy Formalism
N. Banerjee, T. Rahnuma, R. K. Singh
JHEP 10.1007/JHEP04(2023)126

1 · 2022
Asymptotic Symmetry of Four-Dimensional Einstein-Yang-Mills and Einstein-Maxwell Theory
N. Banerjee, T. Rahnuma, R. K. Singh
JHEP 10.1007/JHEP01(2022)033

INVITED TALKS & CONFERENCES

Apr 2026 **Amplitudes, Strong-Field Gravity and Resumption** [INVITED SPEAKER]
Nordita, Stockholm, Sweden
Metric of a Star: A Recursive Multipolar Post-Minkowskian Formalism

Nov 2025 **Integrability, Duality and Related Topics 2025** [INVITED SPEAKER]
APCTP, Pohang, South Korea
Perturbative Scattering Dynamics in AdS

Aug 2025 **2nd APCTP-INPP Demokritos Meeting** [INVITED SPEAKER]
APCTP, Pohang, South Korea
Perturbative Solutions of Einstein's Equations: Recursive Techniques and Multipole Expansions

Dec 2024 **New Frontiers in String and Field Theories 2024** [TALK]
Incheon, South Korea
Studying Gravitational Scattering using Off-shell Recursions

Jan 2024 **APCTP Winter School on Fundamental Physics 2024** [TALK]
Pohang, South Korea
Unified Insights in Gravity: Asymptotic Symmetries in SUGRA using CCFT Techniques

Dec 2023 **Indian Strings Meeting 2023** [POSTER]
IIT Bombay, India
Asymptotic Symmetries in SUGRA using CCFT Techniques

Dec 2023 **10th International Conference on Gravitation and Cosmology (ICGC 2023)** [TALK]
IIT Guwahati, India
Unified Insights in Gravity: Asymptotic Symmetries and Celestial Holography

Jan 2023 **17th Kavli Asian Winter School (KAWS)** [POSTER]
Institute for Basic Science, Daejeon, South Korea
Asymptotic Symmetries from Celestial CFT

Mar 2022 **2022 Chennai–Southampton Workshop**
Southampton University, UK
Holography, gauge theories and black holes.

RESEARCH VISITS

Apr – May 2026 **Niels Bohr International Academy (NBIA)**
Copenhagen, Denmark
Collaborative research on perturbative gravitational theories with Prof. Poul Henrik Damgaard.

Jan 2026 **Gwangju Institute of Science and Technology (GIST)**
Gwangju, South Korea
Collaborative research on AdS_3 and boundary Carrollian holography.

Dec 2025 **Université libre de Bruxelles (ULB)**
Brussels, Belgium
Asymptotic symmetries and algebra at cosmological horizons in dS spacetime.

Sep 2025 **Korea Institute for Advanced Studies (KIAS)** [INVITED SPEAKER]
Seoul, South Korea
Recursive Techniques in Multipolar Post-Minkowskian Formalism

Feb – Mar 2024 **Asia Pacific Center for Theoretical Physics (APCTP)**
Pohang, South Korea · Host: Prof. Junggi Yoon
Seminars delivered:
- Asymptotic symmetries using celestial CFT techniques
- A review on asymptotic symmetries in gauge and gravity theories and the duality

Mar 2024 **Seoul National University (SNU)**
Seoul, South Korea · Host: Prof. Sangmin Lee
Research group visit and seminar presentation.

TEACHING EXPERIENCE

Jan 2020 – 2024

Teaching Assistant, Department of Physics
IISER Bhopal, India

- Mathematical Methods, 3rd year — Dr. Auditya Sharma
- Quantum Field Theory I & II, 4th year — Dr. Arnab Rudra
- General Theory of Relativity, 4th year — Dr. Nabamita Banerjee
- Newtonian and Classical Mechanics, 3rd year — Dr. Nabamita Banerjee
- General Properties of Matter, 2nd year — Dr. Nabamita Banerjee

INTERNSHIPS & PROJECTS

May – Jul 2017

Summer Project, School of Physical Sciences
NISER, Bhubaneswar, India
Measurement of strange B meson lifetime via $B_s^0 \to J/\psi + \phi$ at CMS detector. Monte Carlo simulation and data analysis using ROOT and RooFit.

Apr – Jun 2014

Summer Research Project (DST-INSPIRE)
Utkal University, Bhubaneswar, India
Optical property study of Arsenic Selenide thin film. Characterisation: XRD and UV spectroscopy.

AWARDS & FELLOWSHIPS

2019 – 2024 Joint CSIR-UGC Junior Research Fellowship — IISER Bhopal

Dec 2018 Eligibility for Lectureship (NET) — Joint CSIR-UGC National Eligibility Test

2016 – 2018 IMA Scholarship — Institute of Mathematics and Applications, Bhubaneswar

2013 – 2015 DST-INSPIRE Scholarship (1788/2012) — Dept. of Science & Technology, New Delhi

Jan 2009 Certificate of Merit — Uranium Talent Search Examination

REFERENCES

Prof. Poul Henrik Damgaard
Niels Bohr International Academy, Copenhagen

Prof. Kanghoon Lee
Korea Institute for Advanced Studies, Seoul

Dr. Nabamita Banerjee
Indian Institute of Science Education and Research Bhopal